

ITA0612-Machine Learning

CO4 - AT3 - Correlation and Pattern Analysis

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Question 1 – Combining Customer Transaction and Demographic Data

Datasets

CUSTOMER DATA				
	Customer_ID	Name	Age	City
0	101	Arun	25	Chennai
1	102	Bala	30	Coimbatore
2	103	Charan	28	Madurai
3	104	Deepak	35	Chennai
4	105	Elan	29	Trichy

TRANSACTION DATA			
	Customer_ID	Transaction_ID	Amount
0	101	T001	1500
1	102	T002	2200
2	102	T003	800
3	103	T004	1750
4	104	T005	3000
5	106	T006	1200

Code

```
import pandas as pd

# Load datasets
customers = pd.read_csv("customers.csv")
transactions = pd.read_csv("transactions.csv")

print("CUSTOMER DATA")
print(customers)

print("\nTRANSACTION DATA")
print(transactions)

# Check duplicate Customer IDs in customer data
duplicates = customers[customers.duplicated(subset="Customer_ID",
keep=False)]

print("\nDUPLICATE CUSTOMER IDs:")
```

```

print(duplicates)

# Remove duplicate customer IDs if present
customers = customers.drop_duplicates(subset="Customer_ID")

# Merge both DataFrames
merged_data = pd.merge(
    customers,
    transactions,
    on="Customer_ID",
    how="outer",
    indicator=True
)

# Reset index for consistent indexing
merged_data = merged_data.reset_index(drop=True)

# Identify mismatched records
print("\nMISMATCHED RECORDS:")
print(merged_data[merged_data["_merge"] != "both"])

# Identify missing values
print("\nMISSING VALUES:")
print(merged_data[merged_data.isnull().any(axis=1)])

# Filter valid records
valid_records = merged_data[
    (merged_data["_merge"] == "both") &
    (merged_data["Amount"].notnull())
]

# Group transactions by Customer ID
grouped_data = valid_records.groupby("Customer_ID")["Amount"].sum()

print("\nTOTAL AMOUNT BY CUSTOMER:")
print(grouped_data)

# Sort by Customer ID
sorted_data = valid_records.sort_values(
    by="Customer_ID"
).reset_index(drop=True)

print("\nFINAL SORTED DATA:")
print(sorted_data)

# Check final index
print("\nFINAL INDEX:")
print(sorted_data.index)

```

Duplicate Customer ID Check

```
DUPLICATE CUSTOMER IDs:  
Empty DataFrame  
Columns: [Customer_ID, Name, Age, City]  
Index: []
```

Mismatched and Missing Records

MISMATCHED RECORDS:

```
MISMATCHED RECORDS:  
   Customer_ID  Name  Age  City Transaction_ID  Amount  _merge  
5           105  Elan  29.0  Trichy           NaN      NaN  left_only  
6           106   NaN   NaN   NaN          T006  1200.0  right_only
```

MISSING VALUES:

```
MISSING VALUES:  
   Customer_ID  Name  Age  City Transaction_ID  Amount  _merge  
5           105  Elan  29.0  Trichy           NaN      NaN  left_only  
6           106   NaN   NaN   NaN          T006  1200.0  right_only
```

This demonstrates debugging of unmatched customer IDs and missing values.

Grouped Transaction Data

TOTAL AMOUNT BY CUSTOMER:

```
TOTAL AMOUNT BY CUSTOMER:  
Customer_ID  
101      1500.0  
102      3000.0  
103      1750.0  
104      3000.0  
Name: Amount, dtype: float64
```

This demonstrates `groupby()` and aggregation.

Final Sorted DataFrame

FINAL INDEX:

```
FINAL INDEX:  
RangeIndex(start=0, stop=5, step=1)
```

FINAL SORTED DATA:

FINAL SORTED DATA:

	Customer_ID	Name	Age	City	Transaction_ID	Amount	_merge
0	101	Arun	25.0	Chennai	T001	1500.0	both
1	102	Bala	30.0	Coimbatore	T002	2200.0	both
2	102	Bala	30.0	Coimbatore	T003	800.0	both
3	103	Charan	28.0	Madurai	T004	1750.0	both
4	104	Deepak	35.0	Chennai	T005	3000.0	both

This demonstrates sorting and consistent indexing.

Question 2 – Complete Data Analysis Pipeline

Dataset Files

JANUARY DATA

	Product	Category	Sales	Quantity	City
0	Laptop	Electronics	50000	2	Chennai
1	Phone	Electronics	30000	3	Chennai
2	Chair	Furniture	10000	5	Madurai
3	Table	Furniture	15000	2	Coimbatore
4	Headphones	Electronics	5000	10	Chennai
5	Keyboard	Electronics	8000	4	Chennai
6	Sofa	Furniture	25000	2	Trichy

FEBRUARY DATA

	Product	Category	Sales	Quantity	City
0	Laptop	Electronics	60000	3	Chennai
1	Phone	Electronics	35000	4	Coimbatore
2	Chair	Furniture	12000	6	Madurai
3	Table	Furniture	18000	3	Chennai
4	Headphones	Electronics	7000	12	Trichy
5	Keyboard	Electronics	9000	5	Madurai
6	Sofa	Furniture	28000	2	Chennai

MARCH DATA

	Product	Category	Sales	Quantity	City
0	Laptop	Electronics	55000	2	Coimbatore
1	Phone	Electronics	40000	5	Chennai
2	Chair	Furniture	11000	5	Trichy
3	Table	Furniture	20000	4	Madurai
4	Headphones	Electronics	8500	15	Chennai
5	Keyboard	Electronics	10000	6	Coimbatore
6	Sofa	Furniture	30000	3	Madurai

File Loading and Combined Data

```
sales_january.csv loaded successfully
sales_february.csv loaded successfully
sales_march.csv loaded successfully
```

COMBINED DATA:

	Product	Category	Sales	Quantity	City
0	Laptop	Electronics	50000	2	Chennai
1	Phone	Electronics	30000	3	Chennai
2	Chair	Furniture	10000	5	Madurai
3	Table	Furniture	15000	2	Coimbatore
4	Headphones	Electronics	5000	10	Chennai

Filtered Data

FILTERED DATA:

	Product	Category	Sales	Quantity	City
0	Laptop	Electronics	50000	2	Chennai
1	Phone	Electronics	30000	3	Chennai
3	Table	Furniture	15000	2	Coimbatore
6	Sofa	Furniture	25000	2	Trichy
7	Laptop	Electronics	60000	3	Chennai
8	Phone	Electronics	35000	4	Coimbatore
9	Chair	Furniture	12000	6	Madurai
10	Table	Furniture	18000	3	Chennai
13	Sofa	Furniture	28000	2	Chennai
14	Laptop	Electronics	55000	2	Coimbatore
15	Phone	Electronics	40000	5	Chennai
16	Chair	Furniture	11000	5	Trichy
17	Table	Furniture	20000	4	Madurai
20	Sofa	Furniture	30000	3	Madurai

This demonstrates filtering based on the condition:

```
sales_data["Sales"] > 10000
```

Category-Wise Sales

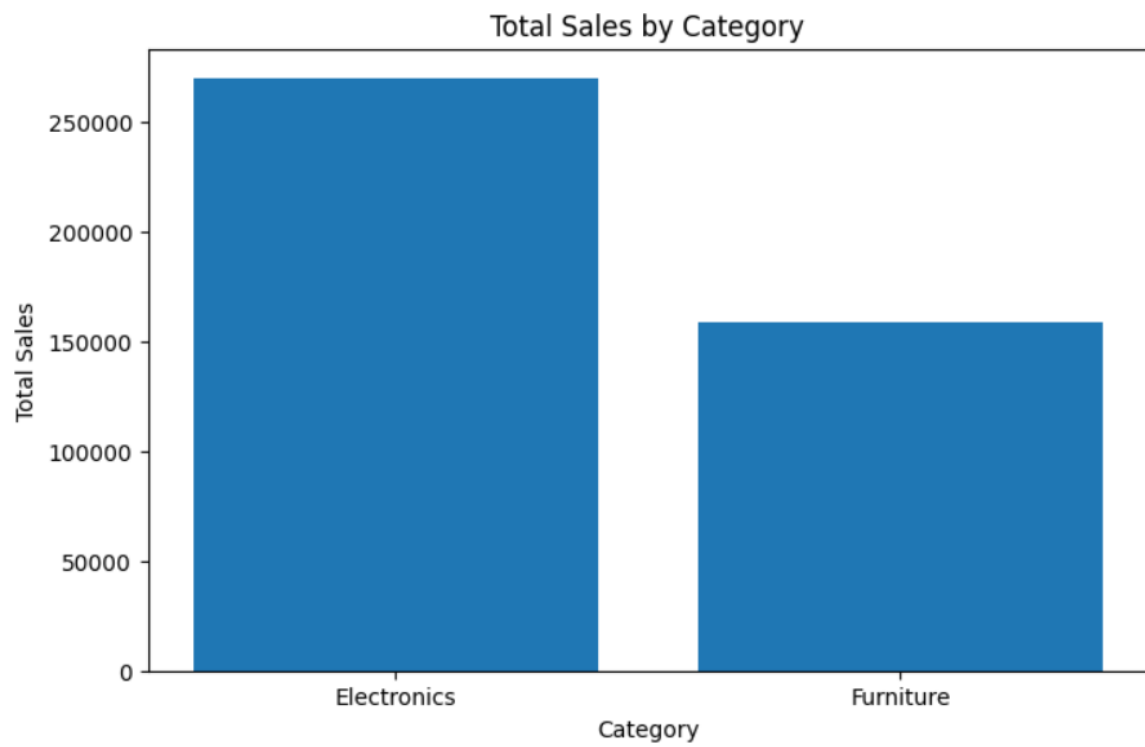
CATEGORY-WISE SALES:

	Category	Sales
0	Electronics	270000
1	Furniture	159000

This demonstrates grouping, aggregation, and sorting.

Bar Chart

Total Sales by Category



Pie Chart

Sales Distribution by Category

